```
d3.mean(data, function (el) {return +el.population });
});
```

Returns the average value of the population attribute of each object in the array, 6856250

Finally, because dealing with minimum and maximum values is a common occurrence, `d3.extent()` conveniently returns `d3.min()` and `d3.max()` in a two-piece array:

```
d3.extent(data, function (el) {return +el.population});
```

Returns [500000, 1300000]

You can also measure nonnumerical data like text by using the JavaScript `.length()` function for strings and arrays. When dealing with topological data, you need more robust mechanisms to measure network structure like centrality and clustering. When dealing with geometric data, you can calculate the area and perimeter of shapes mathematically, which can become rather difficult with complex shapes.

Now that we've loaded, formatted, and measured our data, we can create data visualizations. This requires us to use selections and the functions that come with them, which we'll examine in more detail in the next section.

2.2 Data-binding

We touched on data-binding in chapter 1, but here we'll go into it in more detail, explaining how selections work with data-binding to create elements (the fourth step shown in figure 2.10) and also to change those elements after they've been created. Our first example uses the data from cities.csv. After that we'll see the process using this data as well as simple numerical arrays, and later we'll do more interesting things with the tweets data.

2.2.1 Selections and binding

You use selections to make changes to the structure and appearance of your web page with D3. Remember that a selection consists of one or more elements in the DOM as well as the data, if any, associated with them. You can also create or delete elements using selections, and change the style and content. You've seen how to use `d3.select()` to change a DOM element, and now we'll focus on creating and removing elements based on data. For this example, we'll use cities.csv as our data source, and so we'll need to load cities.csv and trigger our data visualization function in the callback to create a set of new `<div>` elements on the page using this code, with the results shown in figure 2.11.

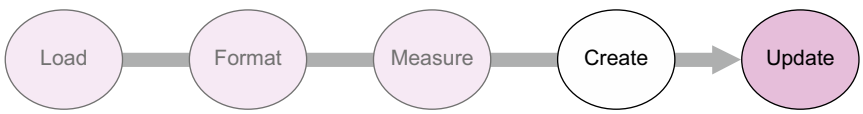


Figure 2.10 To create graphics in D3, you use selections that bind data to DOM elements.